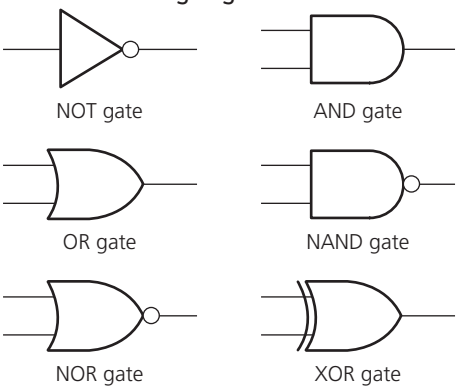
10.1.1 Logic gate symbols

Six different logic gates will be considered in this chapter:



▲ **Figure 10.1** Logic gate symbols

Truth tables

Truth tables are used to trace the output from a logic gate or logic circuit. The NOT gate is the only logic gate with one input; the other five gates have two inputs (see Figure 10.1).

Although each logic gate can only have one or two inputs, the number of inputs to a logic circuit can be more than 2; for example, three inputs give a possible 2^3 (=8) binary combinations. And for four inputs, the number of possible binary combinations is 2^4 (=16). It is clear that the number of possible binary combinations is a multiple of the number 2 in every case. The possible inputs in a truth table can be summarised as shown in Table 10.1.

▼ **Table 10.1** All possible inputs for truth tables with two, three and four inputs

Inputs	
A	B
0	0
0	1
1	0
1	1

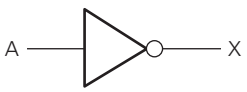
Inputs		
A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Inputs			
A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

As we can see, a truth table will also list the output for every possible combination of inputs.

10.2 The function of the six logic gates

10.2.1 NOT gate



▲ **Figure 10.2**

Description:	Truth table:	How to write this:
The output, X, is 1 if: the input, A, is 0	▼ **Table 10.2** **Input: A** / **Output: X** 0 / 1 1 / 0	X = NOT A (logic notation) X = $\overline{A}$ (Boolean algebra)

Note the use of Boolean algebra to represent logic gates. This is optional at IGCSE but many students may prefer to use this notation (see NOTE later).

10.2.2 AND gate



▲ **Figure 10.3**

Description:	Truth table:	How to write this:
The output, X, is 1 if: both inputs, A and B, are 1	▼ **Table 10.3** **Inputs: A, B** / **Outputs: X** 0 0 0 0 1 0 1 0 0 1 1 1	X = A AND B (logic notation) X = A . B (Boolean algebra)

10.2.3 OR gate



▲ **Figure 10.4**

Description:	Truth table:	How to write this:
The output, X, is 1 if: either input, A or B, or both, are 1	▼ **Table 10.4** **Inputs: A, B** / **Output: X** 0 0 0 0 1 1 1 0 1 1 1 1	X = A OR B (logic notation) X = A + B (Boolean algebra)

10.2.4 NAND gate (NOT AND)



▲ **Figure 10.5**

Description:	Truth table:	How to write this:
The output, X, is 1 if: input A AND input B are NOT both 1	▼ **Table 10.5** **Inputs** / **Output** **A** **B** **X** 0 0 1 0 1 1 1 0 1 1 1 0	X = A NAND B (logic notation) $X = \overline{A \cdot B}$ (Boolean algebra)

10.2.5 NOR gate (NOT OR)



▲ **Figure 10.6**

Description:	Truth table:	How to write this:
The output, X, is 1 if: neither input A nor input B is 1	▼ **Table 10.6** **Inputs** / **Output** **A** **B** **X** 0 0 1 0 1 0 1 0 0 1 1 0	X = A NOR B (logic notation) $X = \overline{A + B}$ (Boolean algebra)

10.2.6 XOR gate



▲ **Figure 10.7**

Description:	Truth table:	How to write this:
The output, X, is 1 if: (input A is 1 AND input B is 0) **or** (input A is 0 AND input B is 1)	▼ **Table 10.7** **Inputs** / **Output** **A** **B** **X** 0 0 0 0 1 1 1 0 1 1 1 0	X = A XOR B (logic notation) $X = (A \cdot \overline{B}) + (\overline{A} \cdot B)$ (Boolean algebra) NOTE: this is sometimes written as: $(A + B) \cdot \overline{(A \cdot B)}$

Activity 10.1

Show why X = (A AND NOT B) OR (NOT A AND B)
and
Y = (A OR B) AND (NOT (A AND B)) both represent the same logic gate.

You will notice in the Boolean algebra, three new symbols; these have the following meaning:

- . represents the AND operation
- + represents the OR operation
- a bar (above the letter or letters, e.g. $\overline{a}$) represents the NOT operation.

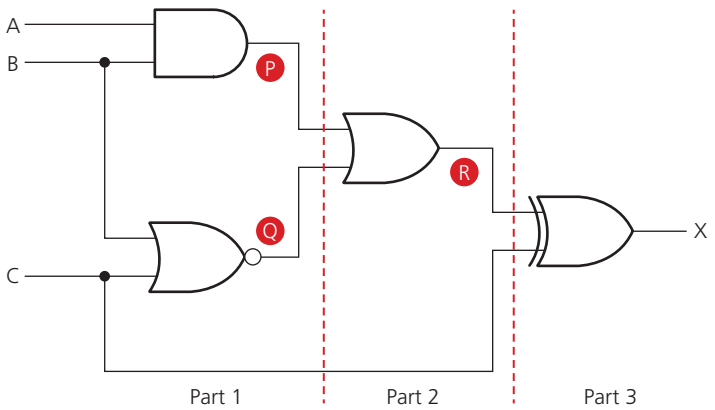
10.3 Logic circuits, logic expressions, truth tables and problem statements

When logic gates are combined together to carry out a particular function, such as controlling a robot, they form a logic circuit. The following eight examples show how to carry out the following tasks:

- Create a logic circuit from a:
  - problem statement (examples 6 and 7)
  - logic or Boolean expression (examples 3 and 8)
  - truth table (examples 4 and 5)
- Complete a truth table from a:
  - problem statement (examples 6 and 7)
  - logic or Boolean expression (examples 3 and 8)
  - logic circuit (example 1)
- Write a logic or Boolean expression from a:
  - problem statement (examples 6 and 7)
  - logic circuit (example 2)
  - truth table (examples 4 and 5).

Example 1

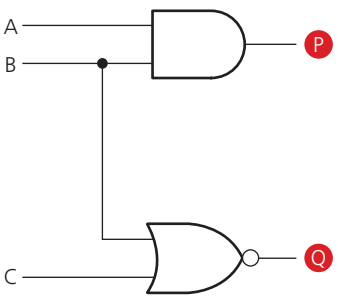
Produce a truth table for the following logic circuit (note the use of black circles at the junctions between wires):



▲ **Figure 10.8**

There are three inputs to this logic circuit, therefore, there will be eight possible binary values that can be input.

To show stepwise how the truth table is produced, the logic circuit has been split up into three parts, as shown by the dotted lines, and intermediate values are shown as P, Q and R.



▲ **Figure 10.9**

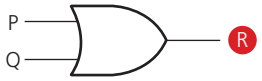
Part 1

This is the first part of the logic circuit; the first task is to find the intermediate values P and Q.

The value of P is found from the AND gate where the inputs are A and B. The value of Q is found from the NOR gate where the inputs are B and C. An intermediate truth table is produced using the logic function descriptions in Section 10.2.

▼ **Table 10.8**

input values			Output values	
A	**B**	**C**	**P**	**Q**
0	0	0	**0**	**1**
0	0	1	**0**	**0**
0	1	0	**0**	**0**
0	1	1	**0**	**0**
1	0	0	**0**	**1**
1	0	1	**0**	**0**
1	1	0	**1**	**0**
1	1	1	**1**	**0**



▲ **Figure 10.10**

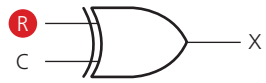
Part 2

The second part of the logic circuit has P and Q as inputs and the intermediate output, R:

This produces the following intermediate truth table. (Note: even though there are only two inputs to the logic gate, we have generated eight binary values in part 1 and these must all be used in this second truth table).

▼ **Table 10.9**

Inputs		Output
P	**Q**	**R**
0	1	**1**
0	0	**0**
0	0	**0**
0	0	**0**
0	1	**1**
0	0	**0**
1	0	**1**
1	0	**1**



▲ **Figure 10.11**

Part 3

The final part of the logic circuit has R and C as inputs and the final output, X:

This gives the third intermediate truth table:

▼ **Table 10.10**

Inputs		Output
R	**C**	**X**
1	0	**1**
0	1	**1**
0	0	**0**
0	1	**1**
1	0	**1**
0	1	**1**
1	0	**1**
1	1	**0**

Putting all three intermediate truth tables together produces the final truth table, which represents the original logic circuit:

▼ **Table 10.11**

Input values			Intermediate values			Output
A	**B**	**C**	**P**	**Q**	**R**	**X**
0	0	0	*0*	*1*	*1*	1
0	0	1	*0*	*0*	*0*	1
0	1	0	*0*	*0*	*0*	0
0	1	1	*0*	*0*	*0*	1
1	0	0	*0*	*1*	*1*	1
1	0	1	*0*	*0*	*0*	1
1	1	0	*1*	*0*	*1*	1
1	1	1	*1*	*0*	*1*	0

The intermediate values can be left out of the final truth table, but it is good practice to leave them in until you become confident about producing the truth tables. The final truth table would then look like this:

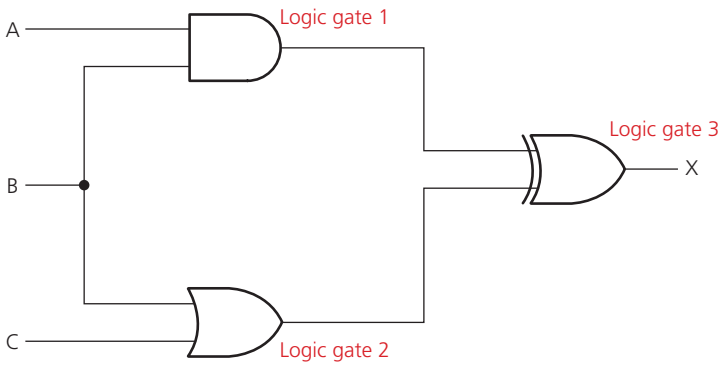
▼ **Table 10.12**

Input values			Output
A	**B**	**C**	**X**
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Example 2

Write logic expressions from the following logic circuits:

a



▲ **Figure 10.12**

The first action is to look at the gates connected to the inputs A, B and C:

logic gate 1: (A AND B)

logic gate 2: (B OR C)

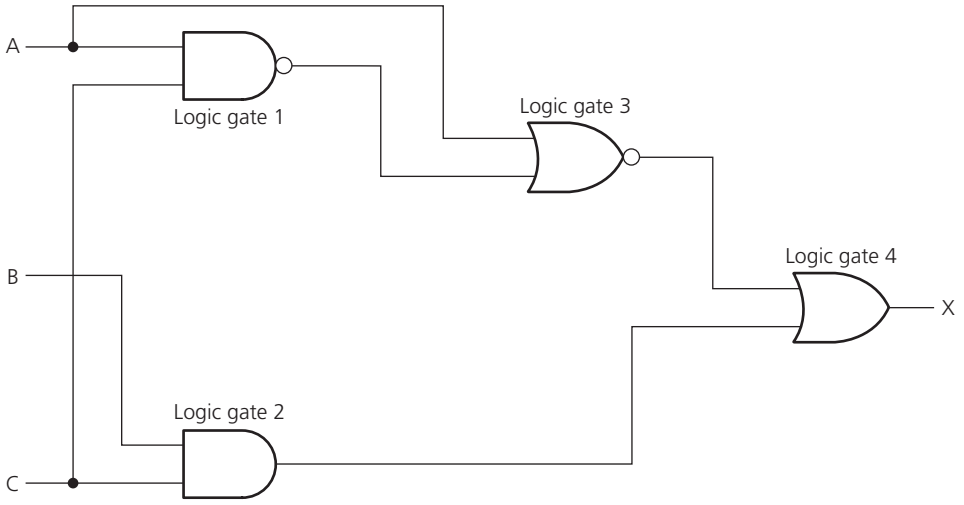
We then join these together using logic gate 3:

[(A AND B)] XOR [(B OR C)] which gives us the required logic expression.

(Note: the square brackets "[ ]" in the expression are not necessary and are used here just for clarity.)

This would be written as: (A AND B) XOR (B OR C)

b



▲ **Figure 10.13**

Again, we will do this in the order of logic gates 1 and 2 first (connected to the three inputs):

logic gate 1: (A NAND C)

logic gate 2: (B AND C)

However, logic gate 3 is also connected to one of the inputs so that should be done next:

logic gate 3: (logic gate 1) NOR A

If we replace (logic gate 1) by the logic expression above, we get:

((A NAND C) NOR A)

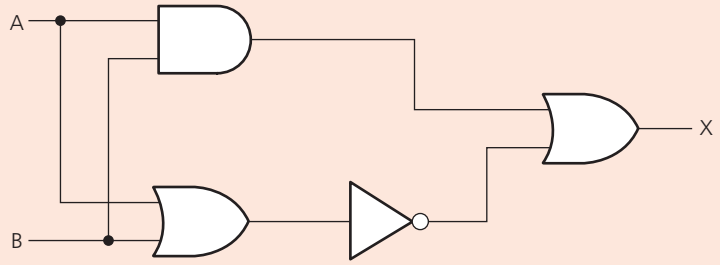
Finally, we can join all these together using:

logic gate 4: ((A NAND C) NOR A) OR (B AND C)

Activity 10.2

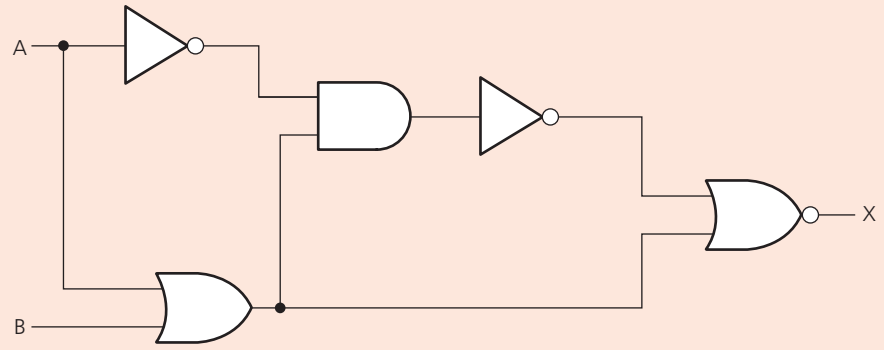
1 Produce truth tables from the following logic circuits:

a



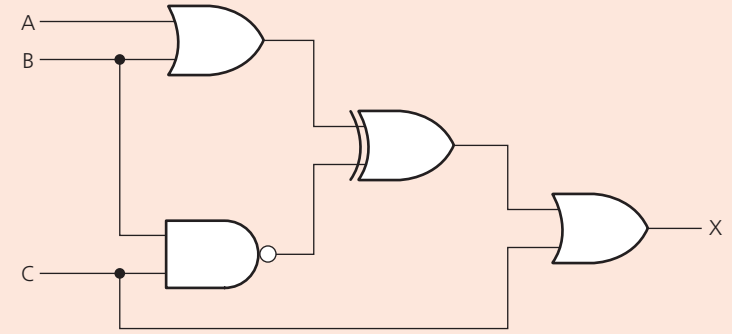
▲ **Figure 10.14**

b



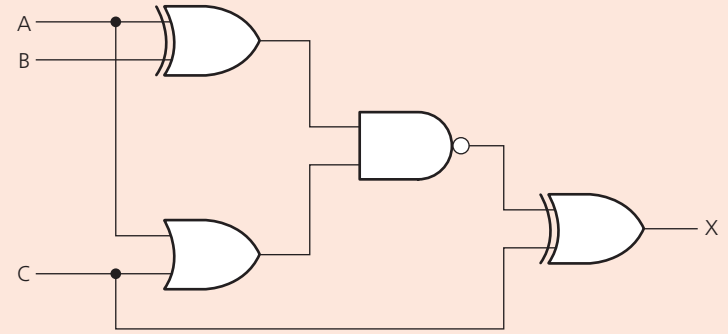
▲ **Figure 10.15**

c



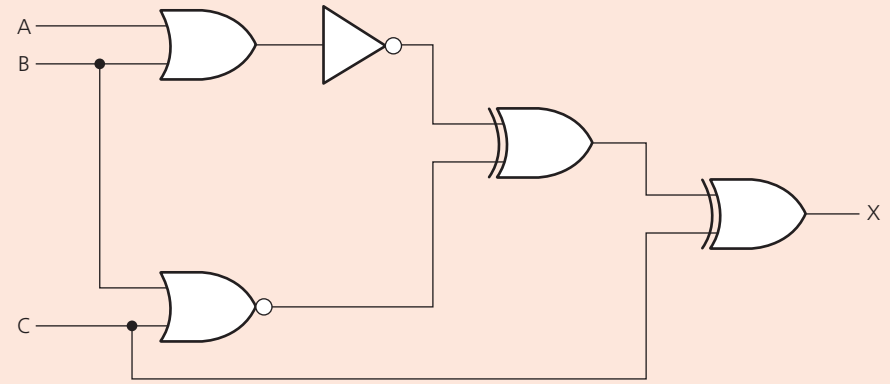
▲ **Figure 10.16**

d



▲ **Figure 10.17**

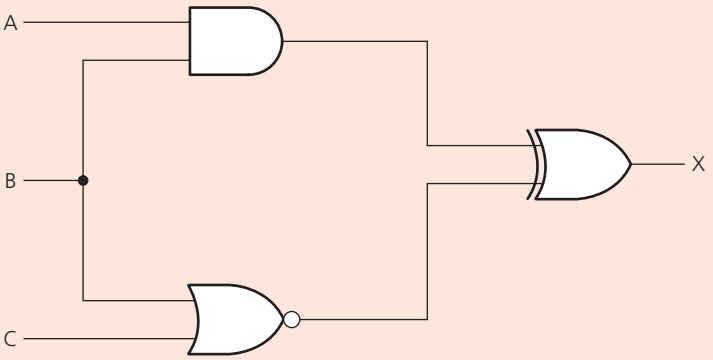
e



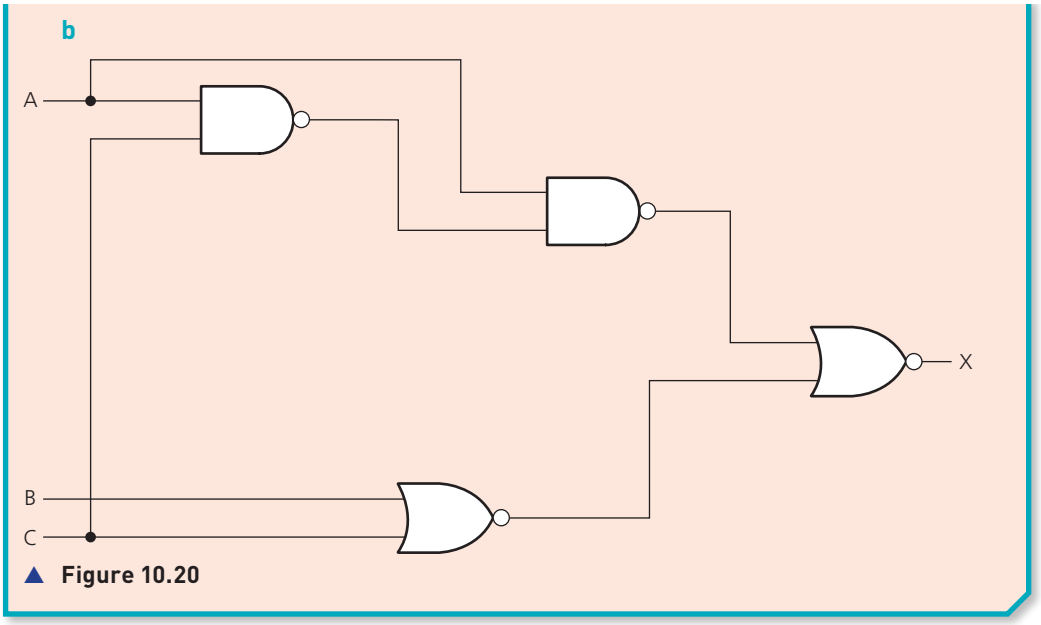
▲ **Figure 10.18**

2 Write logic expressions for the following logic circuits:

a



▲ **Figure 10.19**



▲ **Figure 10.20**

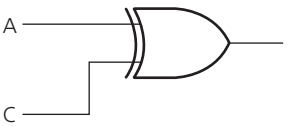
Example 3

A logic circuit can be represented by the following logic expression: (A XOR C) OR (NOT C NAND B)

Produce a logic circuit and a truth table from the above statement.

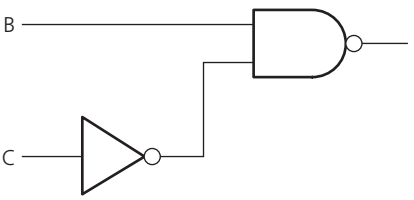
In this example we have a connecting logic gate which is OR.

So, if we produce one half of the circuit from (A XOR C) we get:



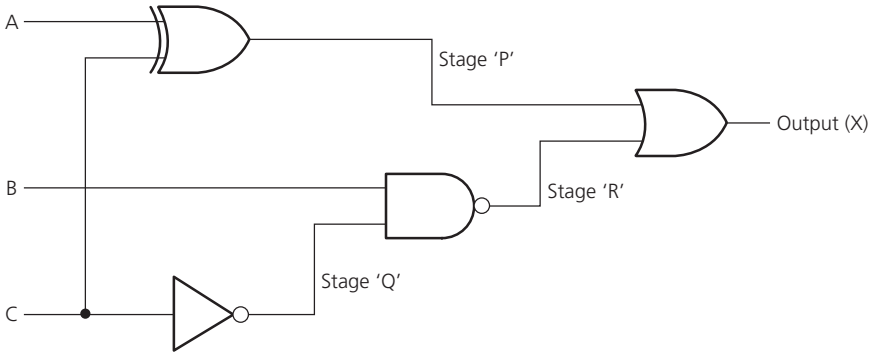
▲ **Figure 10.21**

The other half of the circuit is found from (NOT C NAND B):



▲ **Figure 10.22**

If we now combine these together to form the final logic circuit:



▲ **Figure 10.23**

The truth table is shown:

▼ **Table 10.13**

Input values			Values at stages:			Output
A	**B**	**C**	**'P'**	**'Q'**	**'R'**	**X**
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	0	1	0	0
0	1	1	1	0	1	1
1	0	0	1	1	1	1
1	0	1	0	0	1	1
1	1	0	1	1	0	1
1	1	1	0	0	1	1

Example 4

Look at the two truth tables below; in each case produce a logic expression and the corresponding logic circuit:

a

▼ **Table 10.14**

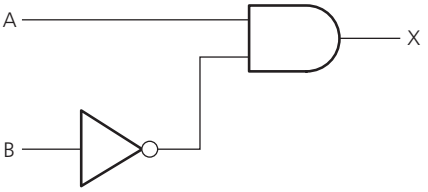
Inputs		Output
A	**B**	**X**
0	0	0
0	1	0
1	0	1
1	1	0

To produce the logic statement, we only concern ourselves with the truth table row where the output value is 1. In this case, A = 1 and B = 0 which gives the logical expression:

A AND NOT B

So we have the logic expression: A AND NOT B

(Note that this could be written as $A \cdot \overline{B}$ in Boolean.) It is now possible to draw the corresponding logic circuit:



▲ **Figure 10.24**

b

▼ **Table 10.15**

Inputs		Output
A	**B**	**X**
0	0	0
0	1	1
1	0	0
1	1	1

This time we have **two** rows where the output is 1; this gives the following logical expressions:

(NOT A AND B)

(A AND B)

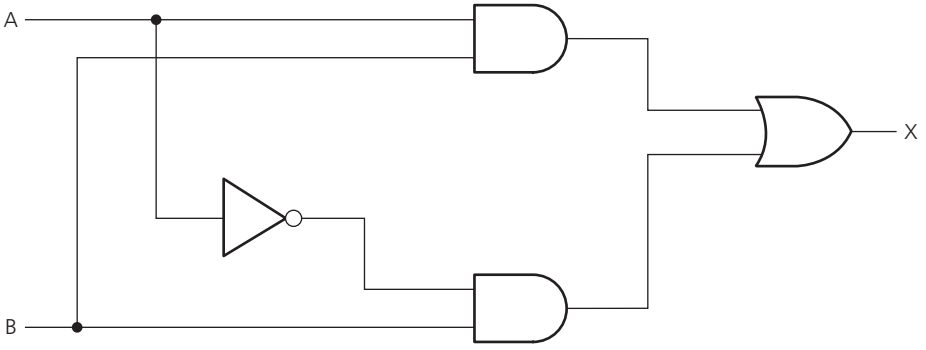
We now join these together using an OR gate to give:

(NOT A AND B) OR (A AND B)

So we have the logic expression: (NOT A AND B) OR (A AND B)

(This could be written as: $\overline{A}.B + A.B$)

It is now possible to draw the corresponding logic circuit:



▲ **Figure 10.25**

Example 5

a Which Boolean expression is represented by the following truth table?

▼ **Table 10.16**

Input values			Output
A	**B**	**C**	**X**
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

We only need to consider those rows where the output is a 1. This gives us the following three logic expressions:

(NOT A AND NOT B AND NOT C)
(A AND NOT B AND NOT C)
(A AND B AND NOT C)

If we now join the three expressions with an OR gate, we end up with the final logic expression:

(NOT A AND NOT B AND NOT C) OR (A AND NOT B AND NOT C) OR (A AND B AND NOT C)

b i Which logic expression is represented by the following truth table?
ii Show that your logic expression in part **i** is the same as: (B AND C) OR (A AND C) OR (A AND B)

▼ **Table 10.17**

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

i We only need to consider those rows where the output is a 1. This gives us the following four Boolean expressions:

(NOT A AND B AND C)
(A AND NOT B AND C)
(A AND B AND NOT C)
(A AND B AND C)

If we now join the four expressions with an OR gate we end up with the following logic expression:

(NOT A AND B AND C) OR (A AND NOT B AND C) OR (A AND B AND NOT C) OR (A AND B AND C)

ii To show that **(B AND C) OR (A AND C) OR (A AND B)** produces the same output as that shown in part **i** we need to produce a new truth table and show that the output is the same as the one in the given truth table:

▼ **Table 10.18**

A	B	C	B AND C	A AND C	A AND B	(B AND C) OR (A AND C) OR (A AND B)	X
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	1	0	1	0	1	1
1	1	0	0	0	1	1	1
1	1	1	1	1	1	1	1

As the second truth table shows, the outputs from logic expressions are both the same; thus the logic expression **(B AND C) OR (A AND C) OR (A AND B)** gives the same output as the logic expression in part **i**.

Activity 10.3

1 Produce:

i a truth table

ii a logic circuit

from the following logic expression:

(NOT A AND B) AND (NOT B OR C)

2 Produce:

i a truth table

ii a logic circuit

from the following logic expression:

(A XOR B) OR ((B NOR C) AND B)

3 Produce:

i a logic expression

ii a logic circuit

from the following truth table:

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

4 Write down a logic expression for **each** of the following truth tables:

a

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

b

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

5 **a** Write down a logic expression corresponding to the following truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

b Show that the following logic expression produces the same output as your answer to part **a** above:

(NOT A AND NOT B) OR (A AND NOT B)